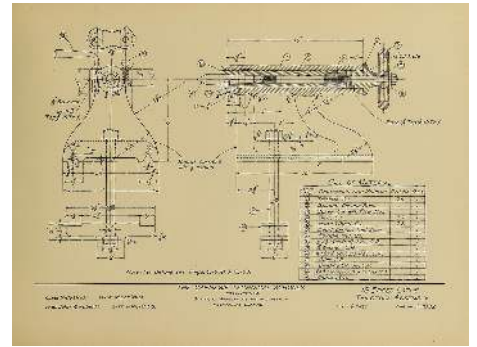
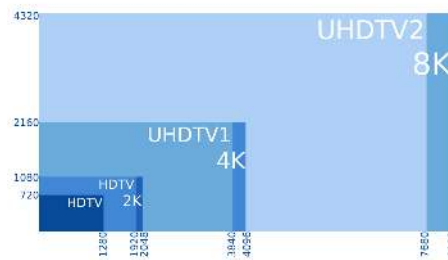




TOPIC OVERVIEW

Topic B — Dilations and Similarity

Scale Factor, Similar Figures, and Area & Perimeter



This overview guide contains...

The topic calendar — every objective and scheduled assessment across this part of the unit

Vocabulary — the terms you must build into your artifacts

Check Your Understanding problem sets — an extra tool to monitor your progress. Completing all questions is not required, but a guide kept up with, completed in full, and turned in at the end of the quarter raises your Standards Mastery Check score by **5%**. Answer keys for every question are posted on the class website.

Since this guide contains so much foundational information vital to success in this course, it is **essential** to keep up with it. It's THE **most important document** needed for success.

Topic Calendar

Monday, August 31st – Friday, September 11th

ARTIFACT 5 · Dilations and Similar Figures

WEEK 4 AUGUST 31–SEPTEMBER 4

BLOCK 1

MON–TUE

AUG 31–SEP 1

I can identify the vertices of an image transformed in a way that does not preserve congruence, when given the vertices or graph of the pre-image. **8.10B**

I can determine the proportional relationship of corresponding sides of similar figures, where the figures have a common side, even with a rotated or reflected image. **8.3A**

I can write a statement about the relationship of the side lengths of a dilated image given the graph of the pre-image. **8.3B**

ARTIFACT 6 · Algebraic Rules of Dilations

ARTIFACT 7 · Area and Perimeter of Dilations

BLOCK 2

WED–THU, SEP 2–3

I can determine the algebraic representation of a dilation when given coordinates and graphs of the pre-image and image, including only being given a graph of the pre-image and the ordered pair of a corresponding point on the graph of the image. **8.3C**

I can identify the ordered pair on the graph of a dilation when given the scale factor as a variable. **8.3C**

I can determine the ratios of area and perimeter of the pre-image and dilated image, including through expressions. **8.10D**

Feedback & Revision Studio · Performance Tasks 1.1 and 1.2

Friday, Sep 4 · also this day: Q1 Mathia Mid-Quarter Check

WEEK 5 SEPTEMBER 7–11

no school Monday

BLOCK 3

TUE–WED, SEP 8–9

Block 2, objectives 1 and 2, continued. **8.3C**

Performance Task 1.3 · Representing Scale Factor Algebraically

in class Tue–Wed, Sep 8–9 · revisions due Friday, Sep 18

BLOCK 4

THU–FRI, SEP 10–11

Block 2, objectives 1 and 2, continued. **8.3C**

Project Write-Up · Improve Dallas

due Friday, Sep 11

AFTER THE TOPIC

Unit 1 Mid-Unit Assessment · all TEKS to date (8.10A–D, 8.3A–C)

Friday, Sep 18 · revisions due Friday, Oct 2

Vocabulary

All of these terms must be eventually added to at least one artifact (Artifact 5, 6, and/or 7!)

pre-image	the original figure (e.g. $ABCD$)
image	the figure after the transformation, named with prime notation (e.g. $A'B'C'D'$)
corresponding sides/ angles/vertices	the matching parts of the pre-image and image
congruent/congruence	same size <i>and</i> shape; a dilation with a scale factor other than 1 is not congruent
rigid motion	a move that keeps the same size and shape (translation, reflection, rotation); a dilation is not a rigid motion
dilation	a transformation that resizes a figure by a scale factor from a center point; it keeps the same shape but changes the size
scale factor	the number every distance from the center is multiplied by; from the origin, $(x, y) \rightarrow (kx, ky)$ – e.g. $k = 2$ doubles, $k = \frac{1}{2}$ halves
center of dilation	the fixed point a figure is resized from (here, usually the origin)
origin	the center of the coordinate plane, point $(0, 0)$; the usual center of dilation
enlargement	a dilation with a scale factor greater than 1 (the image is larger)
reduction	a dilation with a scale factor between 0 and 1 (the image is smaller)
similar figures	figures with the same shape but not necessarily the same size: corresponding angles are equal and corresponding sides are proportional (symbol $\sim$)
similarity	the relationship between two similar figures
proportional	having a constant ratio; in a dilation every side of the image is the same number of times its matching side

ratio	a comparison of two quantities by division, e.g. $A'B' : AB = 2 : 1$
scale model	a real-world enlargement or reduction that keeps the same proportions as the original (e.g. a museum's blue-whale model)
perimeter	the distance around a figure; a dilation by scale factor k multiplies the perimeter by k
area	the amount of surface a figure covers; a dilation by scale factor k multiplies the area by k^2
algebraic representation (rule) of a dilation	the dilation written as mapping notation from the origin: $(x, y) \rightarrow (kx, ky)$
variable scale factor (k)	using a letter k in place of a number so one rule works for any size
expression	a math phrase that may use a variable (e.g. $4x$, k^2); area and perimeter ratios can be written as expressions
vertex/vertices	the corner point(s) of a figure
coordinate plane	a four-quadrant grid with two axes (x-values and y-values) on which transformations of figures are represented

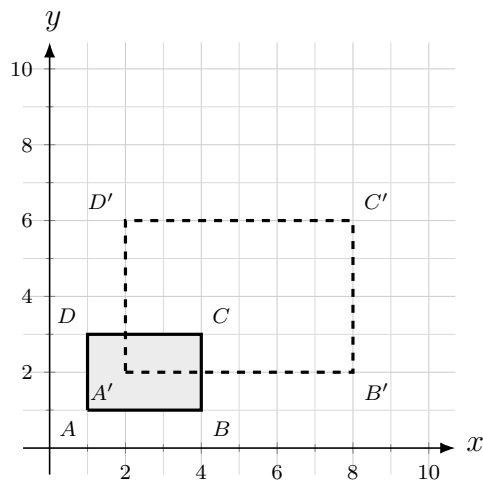
Check Your Understanding

Topic B – Dilations and Similarity

Scale Factor, Similar Figures, and Area & Perimeter

Questions 1-12 cover Artifact 5 (Dilations and Similar Figures)

1. In your own words, describe what a **dilation** does to a figure. Then explain why a dilation by a scale factor of 2 does *not* produce a congruent image.
2. Rectangle $ABCD$ is dilated from the origin to form rectangle $A'B'C'D'$, as shown.



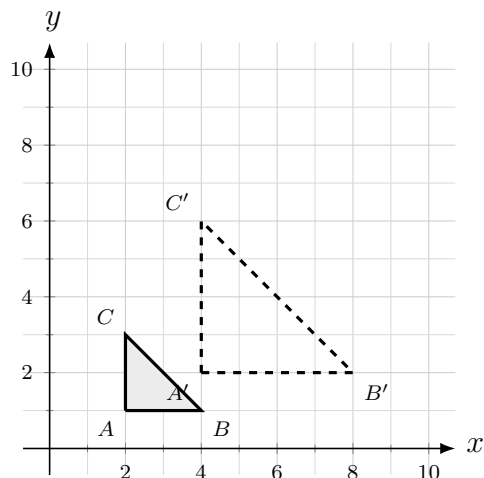
(a) Write the ratio of side $A'B'$ to side AB .

(b) Is the image congruent to the pre-image? Explain how you know.

3. A triangle has side lengths 3, 4, and 5. It is dilated by a scale factor of 3. Write one statement describing how every side length of the image compares to the original, and give the three new side lengths.

4. A figure is dilated by a scale factor of 2 from the origin. Which statement is true?
- (A) The image is congruent to the original figure.
 - (B) Each side length of the image is half the matching side length of the original.
 - (C) The corresponding angle measures stay the same, and each side length doubles.
 - (D) The image has the same perimeter as the original figure.

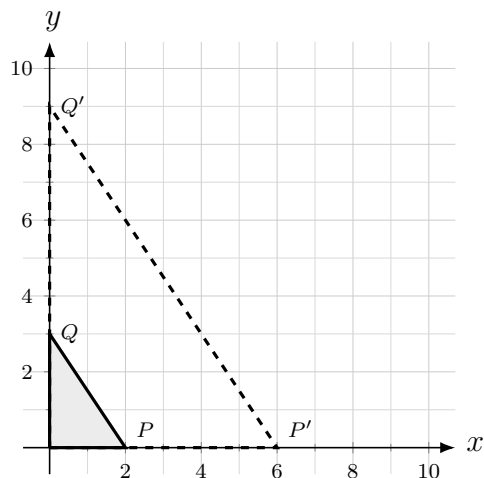
5. Triangle ABC is dilated from the origin to form triangle $A'B'C'$, as shown.



(a) Which vertex of the image corresponds to vertex B , and what are its coordinates?

(b) Explain how you know this transformation does *not* preserve congruence.

6. Two similar triangles share the same vertex at the origin, as shown. The smaller triangle is the pre-image and the larger triangle is its dilation.



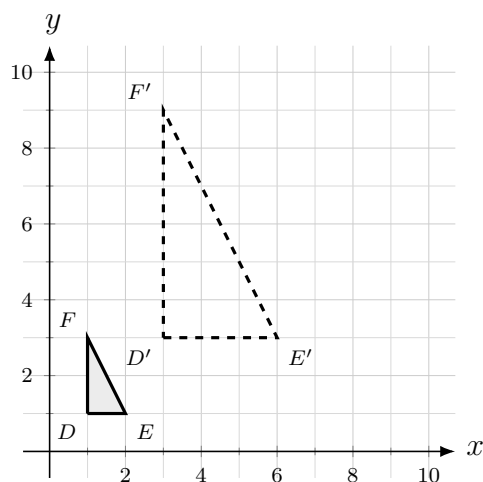
(a) Write the ratio of a side of the larger triangle to its corresponding side of the smaller triangle.

(b) What is the scale factor of the dilation?

7. Two figures are **similar**. Which statement *must* be true?

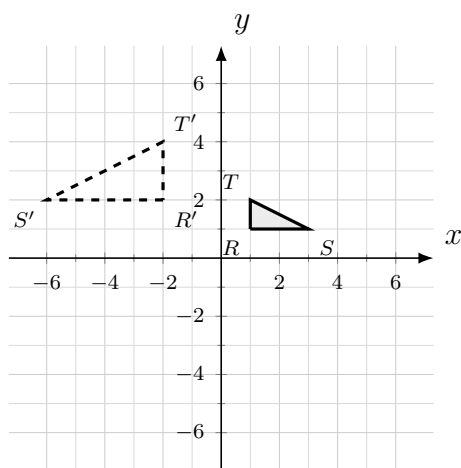
- (A) The two figures are the same size.
- (B) The corresponding angles are equal and the corresponding sides are proportional.
- (C) The corresponding sides are equal in length.
- (D) The two figures have the same perimeter.

8. Triangle DEF is dilated from the origin to form triangle $D'E'F'$, as shown.



Write one statement describing how *every* side length of $D'E'F'$ compares to triangle DEF , and state what that makes the two figures.

9. A figure is dilated from the origin by a scale factor of $\frac{1}{2}$. Is this an **enlargement** or a **reduction**? Describe what happens to the figure and to each of its side lengths.
10. Triangle RST (the pre-image) is reflected across the y -axis and dilated to form the similar triangle $R'S'T'$, as shown. Because the image is flipped, match the sides by correspondence, not by position.



(a) Write the ratio of side $R'S'$ to its corresponding side RS .

(b) Are the two triangles similar, congruent, or both? Explain.

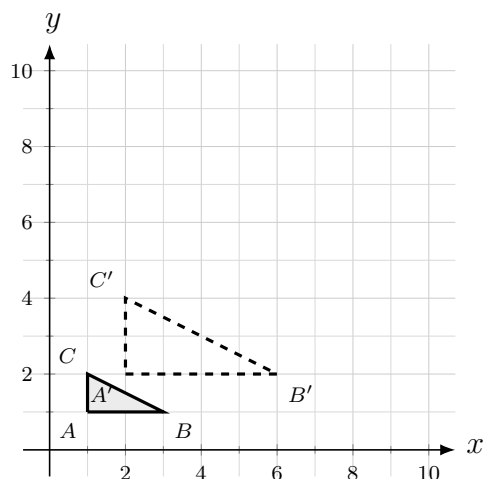
11. Determine whether each transformation produces a **congruent** image or a **similar but not congruent** image. Place a check in the correct column for each row.

	Congruent image	Similar but not congruent
A dilation by a scale factor of 2	☐	☐
A reflection across the x -axis	☐	☐
A dilation by a scale factor of $\frac{1}{2}$	☐	☐
A translation 3 units right and 1 unit up	☐	☐

12. A student claims that two similar figures must be congruent. Decide whether this is correct, and justify with a labeled dilation example that refers to corresponding angles, corresponding sides, and the scale factor.

Questions 13-24 cover Artifact 6 (Algebraic Rules of Dilations)

13. Triangle ABC is dilated from the origin to form triangle $A'B'C'$, as shown. Write the algebraic rule for the dilation on the line beneath the prompt.



Write the algebraic rule $(x, y) \rightarrow (\quad , \quad)$ for this dilation.

$(x, y) \rightarrow (\quad , \quad)$

14. Write the algebraic rule for each dilation from the origin. Write your rule in the parentheses.

(a) scale factor of 2

$(x, y) \rightarrow (\quad , \quad)$

(b) scale factor of 3

$(x, y) \rightarrow (\quad , \quad)$

(c) scale factor of $\frac{1}{2}$

$(x, y) \rightarrow (\quad , \quad)$

(d) scale factor of 5

$(x, y) \rightarrow (\quad , \quad)$

15. A figure is dilated from the origin by a scale factor of 3. Which rule represents this dilation?

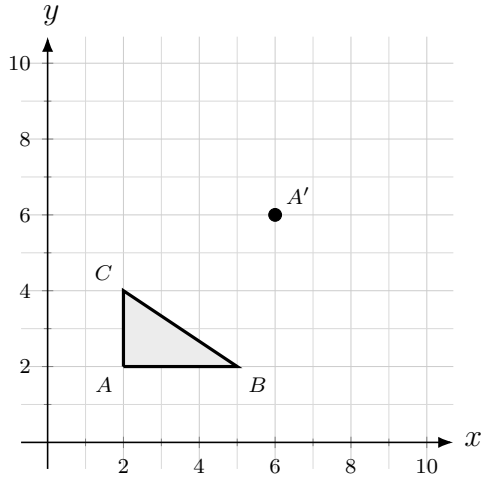
(A) $(x, y) \rightarrow (x + 3, y + 3)$

(B) $(x, y) \rightarrow (3x, 3y)$

(C) $(x, y) \rightarrow (3x, y)$

(D) $(x, y) \rightarrow (x, 3y)$

16. Triangle ABC has vertices $A(2, 2)$, $B(5, 2)$, and $C(2, 4)$. After a dilation from the origin, only one image point was recorded: $A'(6, 6)$, shown as the dot.



(a) Use A and A' to find the scale factor, then write the dilation rule.

$$(x, y) \rightarrow (\quad , \quad)$$

(b) Use your rule to find the coordinates of B' and C' .

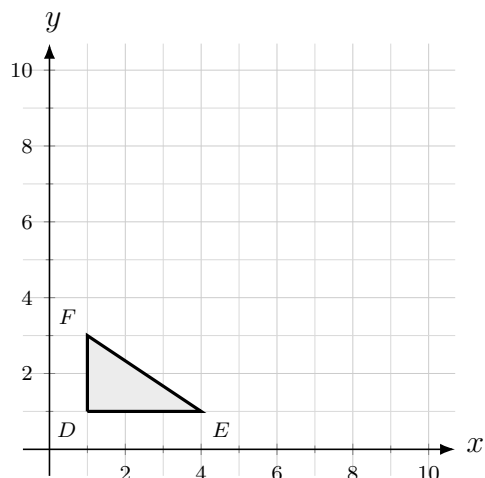
17. A figure is dilated from the origin by the rule $(x, y) \rightarrow (2x, 2y)$. Find the image of each point: $D(3, 5)$ and $E(4, 1)$.

18. A figure is dilated from the origin by a scale factor of k . Write the coordinates of the image of point $P(5, 2)$ in terms of k .

19. A dilation from the origin maps point $Q(3, 4)$ to $Q'(12, 16)$. What is the scale factor, and what is the algebraic rule for the dilation?

20. A part is dilated from the origin by a scale factor of k . A client needs the image of point $B(5, 2)$ to land at $(15, 6)$. What scale factor k produces this, and how do you know?

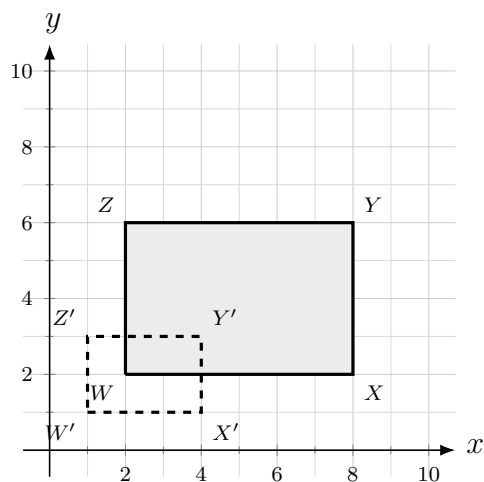
21. Triangle DEF with vertices $D(1, 1)$, $E(4, 1)$, and $F(1, 3)$ is dilated from the origin by a scale factor of 2. Plot the image $D'E'F'$ on the grid below, and label the coordinates of each vertex.



22. Determine whether each rule represents a **dilation**, a **translation**, or a **reflection**. Place a check in the correct column for each row.

	Dilation	Translation	Reflection
$(x, y) \rightarrow (2x, 2y)$	☐	☐	☐
$(x, y) \rightarrow (x + 2, y)$	☐	☐	☐
$(x, y) \rightarrow (-x, y)$	☐	☐	☐
$(x, y) \rightarrow (3x, 3y)$	☐	☐	☐

23. Rectangle $WXYZ$ (the pre-image) is dilated from the origin to form the smaller rectangle $W'X'Y'Z'$, as shown.



(a) Write the algebraic rule for this dilation.

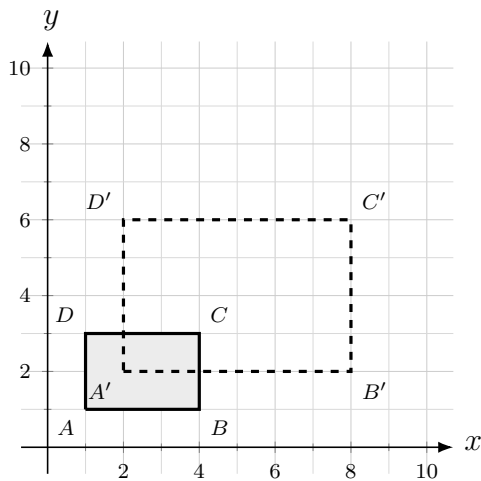
$$(x, y) \rightarrow \left(\quad, \quad \right)$$

(b) Is this dilation an enlargement or a reduction?
How does the rule tell you?

24. Explain why writing a dilation as a rule with a variable scale factor, $(x, y) \rightarrow (kx, ky)$, is more useful than a rule for one fixed size. State what stays the same about a figure's shape for any value of k , using corresponding sides and proportionality.

Questions 25-35 cover Artifact 7 (Area and Perimeter of Dilations)

25. A rectangle is 3 units wide and 2 units tall. It is dilated from the origin by a scale factor of 2.
- (a) Find the perimeter of the original rectangle and of the dilated image, then write the ratio (image : original).
- (b) Find the area of each rectangle and write that ratio (image : original).
26. A figure is dilated from the origin by a scale factor of 3. The area of the image is how many times the area of the original figure?
- (A) 3 times
(B) 6 times
(C) 9 times
(D) 12 times
27. Rectangle $ABCD$ is dilated from the origin by a scale factor of 2 to form $A'B'C'D'$, as shown.



Find the perimeter ratio and the area ratio ($A'B'C'D' : ABCD$). Show the measures you used.

28. Complete the table for a dilation by scale factor k . Write each ratio as image : original. The bottom row uses the variable k .

Scale factor	Perimeter ratio	Area ratio
$k = 2$		
$k = 3$		
$k = 5$		
k (any)		

29. Moving an image from 1080p to a 4K screen dilates it by a scale factor of 2. Explain why the 4K screen has *four* times as many pixels, using how area changes under a dilation.
30. A figure is dilated from the origin by a scale factor of $\frac{1}{2}$. What happens to its area? Write the area ratio (image:original) and explain your reasoning.
31. The area of a dilated image is 16 times the area of the original figure. What scale factor was used? Explain how you found it.
32. A square has a side length of s . It is dilated from the origin by a scale factor of k . Write an expression for the **perimeter** of the image and an expression for the **area** of the image, each in terms of s and k .
33. For a dilation by scale factor k , determine whether each quantity is multiplied by k or by k^2 . Place a check in the correct column for each row.
- | | Multiplied by k | Multiplied by k^2 |
|--|--------------------------|--------------------------|
| The perimeter of the figure | ☐ | ☐ |
| The area of the figure | ☐ | ☐ |
| The length of one side | ☐ | ☐ |
| The distance from the origin to a vertex | ☐ | ☐ |
34. A designer builds a small reference tile and then a version of it dilated by a scale factor of 4. Without computing exact numbers, predict how many times larger the *perimeter* of the big tile is and how many times larger its *area* is. Justify each prediction.

35. Explain why dilating a figure by a scale factor of k multiplies its **perimeter** by k but its **area** by k^2 . Support your answer with a worked numerical example.